

First Hop Redundancy Protocols

At the end of this episode, I will be able to:

1. First Hop Redundancy Protocols (FHRPs)

Leamer Objective: *Learn about the function and variety of First Hop Redundancy Protocols (FHRPs) in use today*

Description: In this episode, you will learn about the need for First Hop Redundancy Protocols (FHRPs) and their function. You will also learn about three of the more popular FHRPs in use today.

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- What if you have multiple routers (default gateways) that can support your hosts? You will need a protocol that can transparently direct traffic to an available router in the group of routers. This is the job of FHRPs.
 - A common concept is a Virtual IP Address (VIP). This is an address that is "assigned" to the group of routers - this enables clients to set a default gateway IP address to the VIP address - by sending traffic to the VIP, the FHRP protocol can decide which router actually receives the traffic.
 - There are three main FHRPs in use today.
 - Hot Standby Router Protocol (HSRP) - this is a Cisco proprietary FHRP that was originally invented in 1998. There are currently two versions with the latter version supporting IPv6. There are two main issues with this protocol today - one is that it is Cisco proprietary, and the other issue is there is no built-in load balancing.
 - Virtual Router Redundancy Protocol (VRRP) - this is basically an open standard version of HSRP. There are currently three versions of this protocol. Like HSRP, the big issue is there is no built-in load balancing.
 - Gateway Load Balancing Protocol (GLBP) - this is another Cisco invention. It provides support for built-in load balancing. In fact, it supports round robin or weighted load balancing configurations.
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Additional Resources:

First Hop Redundancy Protocol

https://en.wikipedia.org/wiki/First-hop_redundancy_protocol